WEB DEVELOPMENT MASTER ROADMAP

A comprehensive 19-phase curriculum covering Frontend, Backend, UI/UX, Databases, APIs, Security, DevOps & Full-Stack Projects.

PHASE 1 Web Development Fundamentals

1. Introduction to Web Development
2. How the Internet & Websites Work
3. Client-Server Architecture
4. Frontend vs Backend & Static vs Dynamic
5. Domain Names & DNS Fundamentals
6. Web Hosting, HTTP/HTTPS & Servers
7. Request & Response Cycle & Browsers
8. Website Deployment Fundamentals

PHASE 2 HTML5

1. HTML Structure, Elements & Tags
2. Headings, Paragraphs & Text Formatting
3. Links, Images, Lists & Tables
4. Forms, Input Types & Validation
5. Semantic HTML & Media (Audio/Video)
6. Iframes & HTML Entities
7. HTML5 APIs & Accessibility (a11y)
8. SEO-Friendly HTML Best Practices

PHASE 3 CSS3

1. CSS Syntax, Selectors, Colors & Backgrounds
2. Typography, Fonts & Box Model
3. Display, Positioning & Z-Index
4. Flexbox & CSS Grid Layouts
5. Pseudo-Classes & Pseudo-Elements
6. Transitions, Transformations & Animations
7. Media Queries & Responsive Web Design
8. CSS Variables & Framework Basics

PHASE 4 JavaScript Fundamentals

1. Syntax, Variables & Data Types
2. Operators, Conditionals & Loops
3. Functions, Arrow Functions & Scope
4. Arrays, Objects & Built-in Methods
5. Hoisting, Closures & ES6+ Features
6. DOM Manipulation, Events & Handling
7. Form Validation & Web Storage (Local/Session)
8. JSON & Error Handling

PHASE 5 Advanced JavaScript

1. Execution Context, Call Stack & Event Loop
2. Asynchronous JS: Callbacks, Promises, Async/Await
3. Fetch API & XMLHttpRequest
4. ES Modules (Import/Export)
5. Destructuring, Spread/Rest, Map & Set
6. Regex, Web/Browser APIs & WebSockets
7. Memory Management & Design Patterns

PHASE 6 Responsive Web Design

1. Mobile-First vs Desktop-First Strategy
2. Fluid Layouts, Flexible Images & Typography
3. CSS Media Queries & Breakpoints
4. Responsive Navigation, Forms & Tables
5. Cross-Browser & Device Compatibility
6. Accessibility-First Design Principles

PHASE 7 UI/UX Design Fundamentals

1. UI/UX Design Principles & User Flow
2. Color Theory & Typography Principles
3. Layout, Visual Hierarchy & Spacing
4. Grid Systems & Wireframing
5. Prototyping & User Research
6. Design Systems & Component Design
7. Figma Fundamentals & Design Patterns

PHASE 8 Frontend Frameworks & Libraries

1. React.js Fundamentals, JSX & Components
2. Props, State Management & React Hooks
3. React Router & Context API
4. Forms, API Integration & Error Boundaries
5. React Performance Optimization
6. Next.js Fundamentals
7. SSR, SSG & CSR Architectures

PHASE 9 CSS Frameworks & UI Libraries

Bootstrap | **Tailwind CSS** | **Material UI** | **Shadcn/UI** | **DaisyUI** | **Sass / SCSS** | **CSS Modules** | **Styled Components** | **Icon Libraries** | **Framer Motion**

PHASE 10 Backend Development Fundamentals

1. Intro to Backend & Server-Side Coding
2. Node.js Fundamentals & Architecture
3. NPM Package Management
4. Express.js Framework, Routing & Middleware
5. Request/Response Handling & REST APIs
6. CRUD Operations & Env Variables
7. Error & File Upload Handling
8. MVC Backend Architecture

PHASE 11 Database Management

1. SQL Basics: MySQL & PostgreSQL
2. Tables, Keys (Primary/Foreign) & Queries
3. Joins, Indexing & Normalization
4. NoSQL Basics: MongoDB Fundamentals
5. MongoDB Collections, Documents & CRUD
6. Database Integration with Backend
7. Database Backup & Recovery Protocols

PHASE 12 API Development & Integration

1. REST API, HTTP Methods & Status Codes
2. API Endpoints, Authentication & Authz
3. API Keys, JWT Auth & OAuth 2.0
4. API Rate Limiting, Validation & Docs
5. Postman & Swagger / OpenAPI
6. GraphQL Fundamentals & WebSocket APIs

PHASE 13 Authentication & Authorization

1. Registration & Login Systems
2. Password Hashing & Reset Workflows
3. Email Verification & Session Management
4. Cookie-Based vs Token-Based (JWT) Auth
5. OAuth 2.0 & Social Login Integration
6. Role-Based Access Control (RBAC) & MFA

PHASE 14 Web Security Fundamentals

1. Web Security Basics & OWASP Top 10
2. SQLi, XSS, CSRF & SSRF Prevention
3. Broken Access Control & Auth Flaws
4. File Upload & Path Traversal Protection
5. CORS, CSP, Security Headers & HTTPS/TLS
6. Input Validation & Output Encoding

PHASE 15 Git & Version Control

1. Git Installation & Global Configuration
2. Repositories, Init, Clone, Add, Commit
3. Push, Pull, Branching & Merging
4. Handling Merge Conflicts, Stash & Rebase
5. GitHub Fundamentals, Pull Requests & Issues
6. GitHub Actions & CI/CD Basics

PHASE 16 Web Hosting & Deployment

1. Domain Registration & DNS Config
2. Shared, VPS & Cloud Hosting
3. Nginx & Apache Web Server Setup
4. Linux Admin & SSL Cert Installation
5. Frontend/Backend Deployment (Vercel, Netlify)
6. Docker & CI/CD Fundamentals

PHASE 17 Testing & Performance Optimization

1. Manual, Unit, Integration & E2E Testing
2. Jest, React Testing Library, Cypress/Playwright
3. DevTools & Debugging Techniques
4. Performance: Lazy Loading, Caching & Code Split
5. Lighthouse, Core Web Vitals & SEO/A11y Testing

PHASE 18 Advanced Web Development Concepts

1. Progressive Web Apps (PWA) & Service Workers
2. Web Push Notifications & Real-Time WebSockets
3. Microservices & Serverless Architectures
4. Payment Gateway & Email Integration
5. Cloud Storage & 3rd Party API Integration
6. Internationalization (i18n)
7. Web Application Scalability
8. Performance & Uptime Monitoring

PHASE 19 Practical Portfolio & Full-Stack Projects

1. Personal Portfolio Website
2. Responsive Business Website
3. High-Converting Landing Page
4. Blog Website with CMS
5. College Management Website
6. Student Management System
7. E-Commerce Website
8. Robust Authentication System
9. Admin Analytics Dashboard
10. Inventory Management System
11. Online Learning Platform
12. Job Portal Platform
13. Real-Time Chat Application
14. URL Shortener Utility
15. Personal Expense Tracker
16. API-Based Weather Application
17. Payment Gateway Integration Project
18. Full-Stack SaaS Application
19. Cybersecurity Academy Website
20. Complete Full-Stack Web Application